Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Multi Observation Production Centre

Global Ocean Surface Carbon Product

MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008

Issue: 7.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated by
0.0	2019/01	All	Creation of the document	Marion Gehlen, Thi Tuyet Trang Chau	Mercator Ocean
1.0	2019/12	All	Update of the description: new methods (reconstruction and uncertainty quantification) and the corresponding validation	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	Mercator Ocean
2.0	2020/09	All	Update of the description and validation results corresponding to changes in: ● input fields in reconstruction of surface ocean partial pressure of carbon dioxide, ● parameterization of surface ocean downward mass flux of carbon dioxide (Wanninkhof 2014 instead of Wanninkhof 1992), uncertainty quantification of surface ocean pH.	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	Stéphanie Guinehut
3.0	2021/09	All	Update of the description and validation results corresponding to extensions of data coverage: ● spatial extension towards the coasts and high latitudes (above 60°N) ● temporal extension for the year 2020	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	Stéphanie Guinehut
3.1	2022/06	all	New template	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	S. Mulet
4.0	2022/09	all	Update of the description and validation results corresponding to: ● update of input data, ● addition of new variables (total alkalinity, dissolved inorganic carbon, saturation state with respect to calcite and aragonite), ● temporal extension for the year 2021, ● changes in units of surface ocean partial pressure of carbon dioxide spco2 (from Pa to µatm) and of surface ocean downward mass flux of carbon dioxide fgco2 (from kg m-2 s-1 to molC m-2 yr-1).	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	S. Mulet

Issue	Date	§	Description of Change	Author	Validated by
5.0	2023/09	all	Update of the description and validation results corresponding to: ● update of input fields, ● increase of spatial resolution (1° to 0.25°), ● temporal extension for the year 2022.	Thi Tuyet Trang Chau, Marion Gehlen, Frédéric Chevallier	H. Etienne
6.10	2024/09	all	First release of the MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 product superseding MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008. Update of the description and validation results corresponding to: ● update of input fields, ● new dataset ● temporal extension for the year 2023, NRT dataset.	Emeric Claudel, Marion Gehlen, Frédéric Chevallier, Emeric Claudel, Thi Tuyet Trang Chau	H. Etienne
7.0	2025/11	all	MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008. Update of the description and validation results corresponding to: ● update of input fields, ● temporal extension for the year 2024, ● change name from MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 to MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008	Aude Barton, Marion Gehlen, Frédéric Chevallier, Thi Tuyet Trang Chau	H. Etienne

ACRONYM TABLE

Acronym	Description
Argo	Array for Real-time Geostrophic Oceanography; International profiling float network
DIC	Dissolved Inorganic Carbon
EAN	Estimated Accuracy Number
MY	Multi-year
NRT	Near Real Time
REP	Reprocessing
TAC	Copernicus Marine Service Thematic Assembly Centre
REP	Reprocessing
SLA	Sea Level Anomaly
SSS	Sea Surface Salinity
SST	Sea Surface Temperature

KEY DEFINITIONS

Variables	Definition
fgco2	Surface ocean downward mass flux of carbon dioxide expressed as carbon (positive for flux into the ocean)
omega_ar	Saturation state of surface seawater with respect to aragonite
omega_ca	Saturation state of surface seawater with respect to calcite
pH	Surface pH on total scale
spco2	Surface ocean partial pressure of carbon dioxide
talk	Total alkalinity in surface seawater
tco2	Surface ocean Dissolved Inorganic Carbon (DIC)

RELEVANT DOCUMENT LIST

CMEMS-LSCE	Chau, T.-T.-T., Gehlen, M., Metzl, N., and Chevallier, F.: CMEMS-LSCE: a global, 0.25°, monthly recon-struction of the surface ocean carbonate system, Earth Syst. Sci. Data, 16, 121–160, https://doi.org/10.5194/essd-16-121-2024, 2024b.

TABLE OF CONTENTS

ACRONYM TABLE4

KEY DEFINITIONS4

RELEVANT DOCUMENT LIST5

I EXECUTIVE SUMMARY7

I.1 7

I.2 7

I.3 10

II 11

II.1 11

II.2 13

III 15

III.1 15

III.2 15

III.3 16

IV 16

IV.1 17

IV.2 20

IV.3 22

V 28

VI 29

VII 30

I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document presents the approaches and tools used to produce and validate the Global Ocean Surface Carbon Product **MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008**.

Short Description	Product code	Area	Delivery Time
Surface Carbon	MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008	Global	Yearly and monthly

The product provides a dataset (**cmems_obs-mob_glo_bgc-car_my_na_irr-i**) of ocean surface carbon variables on a regular grid (0.25°×0.25°) with a monthly resolution from 1985 until 2024. These variables are

- spco2 - surface ocean partial pressure of carbon dioxide,
- fgco2 - surface ocean downward mass flux of carbon dioxide expressed as carbon (positive for flux into the ocean),
- ph - surface pH on total scale,
- talk - total alkalinity in surface seawater,
- tco2 - surface ocean Dissolved Inorganic Carbon (DIC),
- omega_ca - saturation state of surface seawater with respect to calcite, and
- omega_ar - saturation state of surface seawater with respect to aragonite.

The product also covers the months after December 2024 for spco2 and fgco2 with a dataset updated each month with the data of the previous month (**cmems_obs-mob_glo_bgc-car_nrt_na_irr-i**).

I.2 Summary of the results

Surface ocean partial pressure of carbon dioxide **(spco2 [µatm])**

Surface ocean partial pressure of CO_2 is obtained from an ensemble of Feed Forward Neural Networks (FFNNs) which is referred to as CMEMS-LSCE-FFNNv2 (Chau et al., 2022; 2024a; 2024b). The models were trained on 100 subsampled datasets from the Surface Ocean CO_2 Atlas (SOCAT[1]). Sea surface salinity, temperature, sea surface height, mixed layer depth, atmospheric CO_2 mole fraction, chlorophyll, spco2 climatology, latitude and longitude are used as predictors (Table in Sec. II).

The CMEMS-LSCE-FFNNv2 approach follows a leave-***p***-out cross-validation (***p*** is the number of spco2 observations corresponding to the month of reconstruction are excluded from the training data), allowing the quality of the reconstruction to be assessed against independent SOCAT data. At the global scale and over the full period of reconstruction (1985-2024) the root-mean-square difference (RMSD) is 20 µatm (open ocean: 13.9 µatm, coastal: 26.7 µatm) and the coefficient of determination (r^2) is 0.77 (open

[1] https://www.socat.info/, last access: 13/10/2025

ocean: 0.84, coastal: 0.72). CMEMS-LSCE-FFNNv2 quantifies spatial and temporal model uncertainties in terms of standard deviations of the 100-member ensembles (see Sec. II.2 and Sec. 0).

The SOCAT database is an extensive compilation of measurements of near-surface CO_2 fugacity made from a variety of observing platforms and instruments (Bakker et al., 2016). The period covered by the reconstruction corresponds to a percentage larger than 85% of that data set. As shown in **Error! Reference source not found. Error! Reference source not found.**, the spatial distribution of observations is, however, very heterogeneous and the data density remains poor over large areas of the global ocean (e.g., large areas of the Pacific Ocean and of the Indian Ocean) or seasonally biased (high latitudes). The approach ensures an optimal usage of available data but remains sensitive to data density.

Surface ocean downward mass flux of carbon dioxide **(fgco2 [molC m^{-2} yr^{-1}])**

The flux of CO_2 fgco2 was estimated using the gas exchange formulation

$$fgco2 = -k\rho L(1 - fseaice)(spco2 - apco2) \qquad \text{Eq. (1)},$$

where k is the gas transfer velocity (or piston velocity), ρ is the seawater density, L is the temperature-dependent solubility of CO_2 (Weiss, 1974), $f_{\text{sea ice}}$ is the sea ice fraction, and apco2 is the atmospheric partial pressure of CO_2.

The global ocean CO_2 sink corresponds to the global integral of fgco2 over space and time. In 2024 [resp. 2023], the global ocean CO_2 sink was 2.47±0.13 [resp. 2.58±0.13] PgC/yr (Figure 1 and Table 4). The number behind the ± sign corresponds to the standard deviation (σ) computed from the ensemble and it is thus an estimate of the uncertainty associated with the reconstruction. The average over the full period 1985-2024 is 1.68±0.10 PgC/yr with an interannual variability (temporal standard deviation) of 0.52 PgC/yr. Taking into account the total ocean area of 361.9e6 km^2 and the outgassing of river carbon of 0.65 PgC/yr (Regnier et al., 2022) yields an anthropogenic carbon uptake by the ocean of 3.12±0.32 PgC/yr [3.23±0.33 PgC/yr] for 2024 [2023], and 2.33±0.32 PgC/yr averaged over the years 1985-2024. Our data-based estimates of the anthropogenic fluxes are in line with the estimates in Global Carbon Budget 2024 (GCB2024) by Friedlingstein et al. (2025) (see Table 4).

Surface ocean pH on total scale **(ph [-]),** ***total alkalinity*** **(talk [µmol kg^{-1}]),** ***dissolved inorganic carbon*** **(tco2 [µmol kg^{-1}]), and** ***saturation state with respect to calcite*** **(omega_ca [-])** ***and to aragonite*** **(omega_ar [-])**

Time- and space-varying alkalinity fields were obtained from a multivariate linear regression (LIAR, Carter et al., 2016, 2018) with salinity, temperature, dissolved silica and nitrate (Table , Sec. II) used as independent variables. Surface ocean pH, Dissolved Inorganic Carbon (DIC), and saturation state of seawater with respect to calcite and aragonite were computed from the reconstructed spco2, and reconstructed surface ocean alkalinity using the CO2sys speciation software (Van Heuven et al., 2011; Lewis and Wallace, 1998). Uncertainty of pH, total alkalinity, DIC, and saturation state at each grid cell and time step is derived from LIAR and CO2sys error propagation of their input fields' uncertainties.

The reconstructed alkalinity, DIC, and pH have been assessed with GLODAPv2.2023 bottle data. This product provides direct measurements of alkalinity, DIC, and pH.

For the period 1985-2021, CMEMS-LSCE results in open-ocean alkalinity [DIC] estimates with an RMSD of 14.93 [16.44] µmol kg^{-1}, and an r^2 value of 0.93 [0.95]. The coastal reconstruction of these two variables is much less skillful than the open ocean (Table 1). The global error between pH from GLODAPv2.2023 bottle data and the reconstructed pH over the period 1985-2021 is 0.024 pH units for RMSD (open: 0.021, coastal: 0.038), and the r^2 value is 0.68 (open: 0.72, coastal: 0.51). pH decreases in

response to the uptake of CO_2 by the ocean. The rate of decrease computed over the period of reconstruction is 0.017±0.019 pH units per decade (Figure 1).

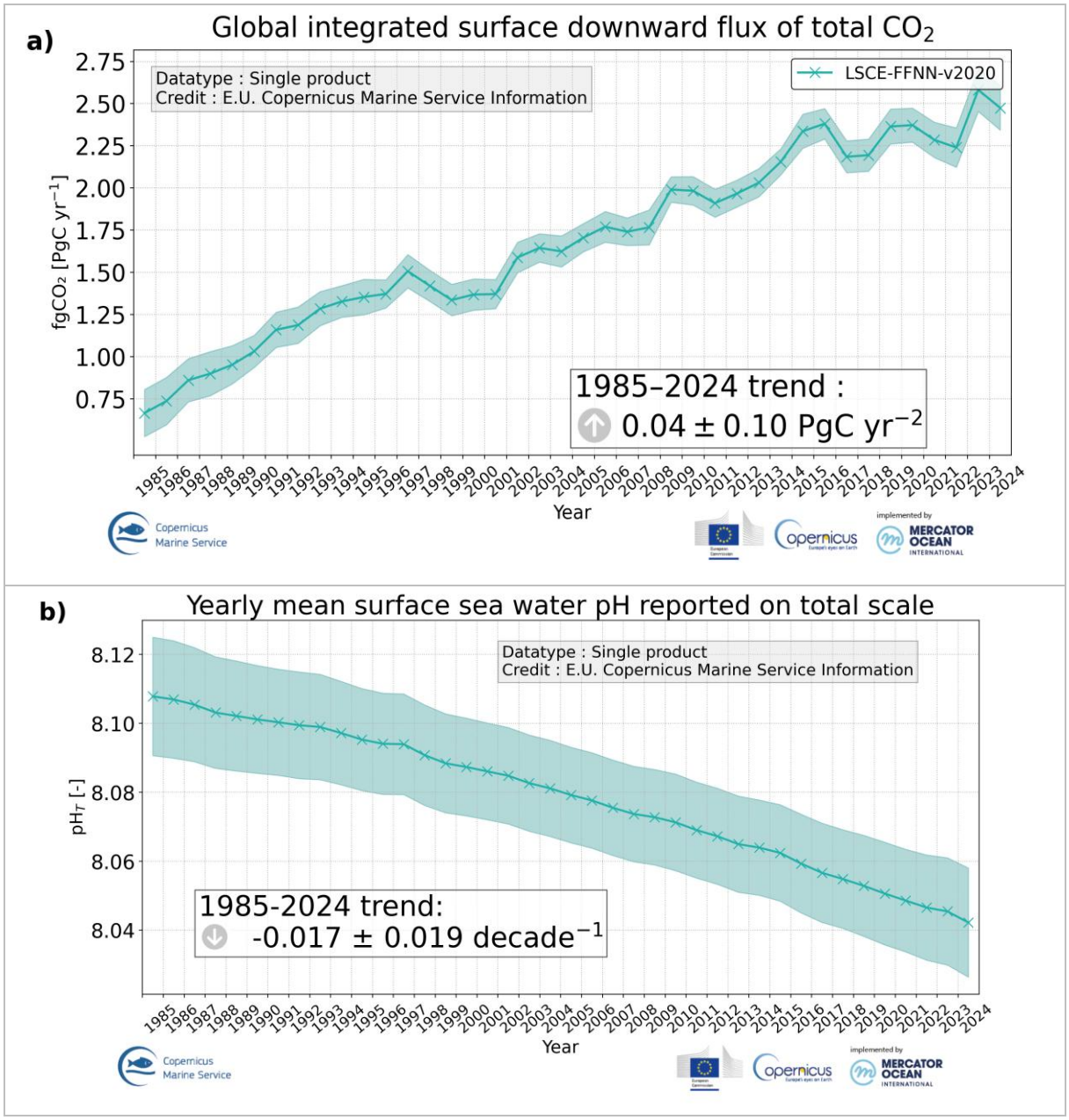

Figure 1: (a) Area integrated yearly surface downward flux of total CO_2 for the period from 1985 to 2024 (GLOBAL_OMI_HEALTH_CARBON_co2_flux_integrated product). (b) Yearly mean surface pH reported on total scale (GLOBAL_OMI_HEALTH_CARBON_ph_area_averaged CMEMS product). The uncertainty envelope is defined as the 68% confidence interval (estimate ± uncertainty) of yearly means. Trend and uncertainty are defined as the slope and its residual standard deviation estimated with a linear least-squares regression.

I.3 Estimated Accuracy Numbers

The quality of the reconstruction is assessed against independent SOCAT data in a leave-*p*-out cross-validation where *p* spco2 observations corresponding to a given month of reconstruction have been excluded from the training data. A ratio of 2:1 for training and validation is used. At the global scale and over the full period of reconstruction (1985-2024), the main statistics are reported in Table 1.

For total alkalinity, DIC, and pH, the statistics are with respect to the independent GLODAPv2.2023 bottle data over the period 1985-2021.

Table 1: Estimated skill over the open (O), and coastal (C) oceans. MAD = Mean Absolute Deviation, RMSD = Root Mean Square Deviation, r^2 = coefficient of determination (see section III for definitions).

Variable		Metrics			Units
		MAD	RMSD	r^2	
spco2	(O)	9.45	13.89	0.84	µatm
	(C)	16.47	26.69	0.72	
talk	(O)	9.76	20.56	0.90	µmol kg^{-1}
	(C)	27.23	56.57	0.83	
tco2	(O)	11.37	21.12	0.92	µmol kg^{-1}
	(C)	26.10	51.68	0.75	
ph	(O)	0.013	0.021	0.71	pH total scale
	(C)	0.023	0.037	0.51	

II PRODUCTION SYSTEM DESCRIPTION

Production centre description for the version covered by this document.

The data have been produced by LSCE using computing facilities provided by the IPSL Spirit cluster and by the HPC resources of TGCC under the allocation A0170102201 made by GENCI.

Production centres name: Laboratoire des Sciences du Climat et de l'Environnement (LSCE) (MULTIOBS-LSCE-GIF-FR)

Production system name: CMEMS-LSCE-FFNNv2

II.1 Origin of input fields

In addition to the predictors listed in Table , latitude and longitude are also used as predictors for the reconstruction of surface ocean pCO_2.

Table 2: Detail of the input data

Reconstruction and prediction of surface ocean CO_2 partial pressure		
Variable name	**References**	**Source**
Target data: surface ocean CO_2 partial pressure		
Measurements of CO_2 fugacity	Bakker et al., 2016; 2025	SOCAT 2025 (last access 20/06/2025)
Predictor data		
Sea surface temperature	Good et al. 2020	CMEMS: SST_GLO_SST_L4_REP_OBSERVATIONS_010_011 (1985-2021) SST_GLO_SST_L4_REP_OBSERVATIONS_010_011 (1985-May 2022) SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001 (June 2022-present)
Sea surface salinity	Buongiorno Nardelli et al. 2016; Droghei et al. 2016, 2018	CMEMS: MULTIOBS_GLO_PHY_S_SURFACE_MYNRT_015_013 (1993-present) Climatology (1985-1992) computed from the available data (1993-2024)
Sea surface height	CLS-TOULOUSE	CMEMS: SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046 (2023SEALEVEL_GLO_PHY_L4_MY_008_047; 1993- May 2024) SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046 (June 2024-present) Climatology plus linear trend (1985-1992) computed from the available data (1993-2024)
Mixed layer depth	Menemenlis et al., 2008	ECCO2: "Estimating the Circulation and Climate of the Ocean" project Phase II (1992-2024) Climatology (1985-1991 or 2025-present) computed from the available data (1992-1997 or 2020-2024)

Variable name	References	Source
Chlorophyll	GlobColour	CMEMS:OCEANCOLOUR_GLO_BGC_L4_MY_009_104 (1998-present) nicus.eu/product/OCEANCOLOUR_GLO_BGC_L4_MY_009_104/description Climatology computed from the available data (1985-1997)

Reconstruction and prediction of surface ocean CO_2 partial pressure		
Variable name	**References**	**Source**
Predictor data		
Atmospheric CO_2 mole fraction	Chevallier et al. 2005, 2010; Chevallier, 2013	CO_2 atmospheric inversion from the Copernicus Atmosphere Monitoring Service Surface: v23r1 (1985-2023) Satellite: FT24r3 (2024-present) Surface Measurements (2025-present): NOAA
spco2 climatology	Takahashi et al. (2009)	LDEO spco2 climatology

Reconstruction of surface ocean downward mass flux of CO_2		
Variable name	**References**	**Source**
6-hourly wind speed Total pressure	Hersbach et al. 2018	ERA5 hourly data on single levels from 1959 to present
Atmospheric CO2 mole fraction	Chevallier et al. 2005, 2010; Chevallier, 2013	CO_2 atmospheric inversion from the Copernicus Atmosphere Monitoring Service (
Sea ice fraction	Good et al. 2020	CMEMS SST_GLO_SST_L4_REP_OBSERVATIONS_010_011 (1985-May 2022) SST_GLO_SST_L4_NRT_OBSERVATIONS_010_001 (June 2022-present)

Reconstruction of surface ocean pH on total scale		
Variable name	**References**	**Source**
Nitrate Silicate Phosphate	Garcia et al. 2019	World Ocean Atlas 2018
Alkalinity, DIC, pH (validation data)	Lauvset et al 2023	GLODAPv2.2023 bottle dataGLODAPv2.2023 bottle data

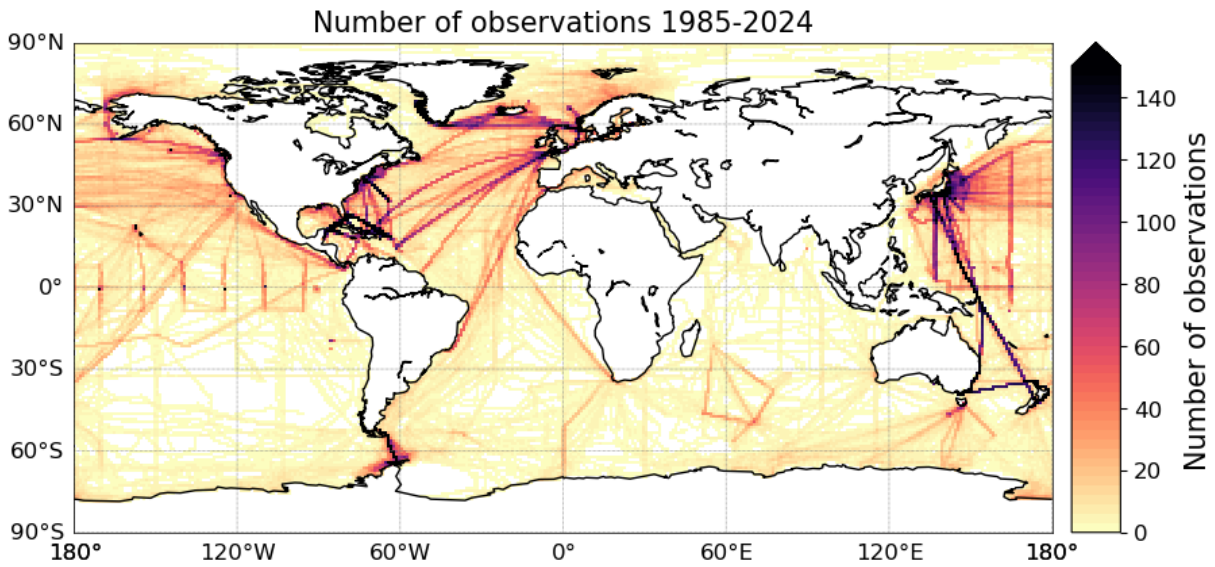

Figure 2: Distribution of SOCAT v2025 – ship traces for the period Jan 1985-Dec 2024 (Bakker et al., 2025): amount of monthly data on 0.25°×0.25° regular grid. A combination of the original 0.25°-coastal-ocean datasets and the 1°-open-ocean datasets downscaled to a resolution of 0.25°×0.25° is used in model fitting (see Sec. II.2).

II.2 Description of the system

(a) Reconstruction of surface ocean carbon dioxide partial pressure

Central to the production process is the reconstruction of surface ocean carbon dioxide partial pressure (spco2) by a Feed Forward Neural Network (FFNN) model following an ensemble-based approach (Chau et al. 2022, 2024a, 2024b). The ensemble network was developed at the Laboratoire des Sciences du Climat et de l'Environnement (LSCE) and validated as an improvement of LSCE-FFNN-v1 by Denvil-Sommer et al. (2019). See Chau et al. (2020) for intercomparison. The model outputs are part of the Copernicus Marine Environment Monitoring Service (CMEMS). It is hence referred to as CMEMS-LSCE-FFNNv2.

Taking advantage of the Surface Ocean CO2 Atlas (SOCAT) (Bakker et al., 2016), the ensemble-based FFNN approach was first used to reconstruct the global distribution of spco2 with a monthly resolution over the global 1°×1° gridded surface ocean (Chau et al, 2022). In 2023, an extension proposed by Chau et al. (2023) was adopted to map spco2 to the global surface ocean with a 16-fold increase in spatial resolution. SOCAT provides the two monthly, gridded products: one covers the global ocean at 1° spatial resolution, and another includes coastal data on 0.25° grid cells. To train the FFNNs, we create a global dataset of spco2 at monthly, 0.25° resolutions (Chau et al., 2024) by combining the original coastal SOCAT dataset and a 0.25° open-ocean dataset. The latter is a downscaled version of the 1° open-ocean product wherein its data are duplicated conservatively for sixteen 0.25° subgrid cells.

The FFNN models were trained on 100 datasets sub-sampled randomly from the drivers and SOCAT data with a ratio of 2:1 for training and validation. The model is sensitive to observation coverage. This limitation is partly overcome by the FFNN approach as the reconstruction of monthly global ocean distributions draws on a larger data set such that FFNN outputs remain close to realistic values. The approach

further uses observations in a 3-month gliding window centered on the month of reconstruction. Note that the data from the month of reconstruction is excluded from the training data in order to reduce model overfitting.

The means (μ) and standard deviations (σ) are computed from the 100-member ensembles of surface ocean partial pressure of CO2 reconstructions, as well as surface ocean downward mass flux of CO2. The standard deviation stands for the associated model uncertainty. In this report, the notation μ±σ expresses a best estimate of the surface ocean partial pressure of CO_2 along with its relative uncertainty. An identical ensemble-based approach is applied to the other carbon variables (pH, total alkalinity, dissolved inorganic carbon, and saturation state with respect to calcite and aragonite). For the uncertainty of a temporal mean of the carbon variables (e.g., **Error! Reference source not found.**, **Error! Reference source not found.**, **Error! Reference source not found.**d **Error! Reference source not found.**), we define its representative estimate as the standard deviation of the difference between the temporal mean from the ensemble members and the one from the ensemble mean.

(b) Reconstruction of surface ocean downward mass flux of carbon dioxide expressed as carbon

The surface ocean downward mass flux of CO_2 is computed from the gas exchange formulation (Eq. 1).

(c) Reconstruction of surface ocean pH, total alkalinity, dissolved inorganic carbon, and saturation state with respect to calcite and to aragonite.

Time- and space-varying surface ocean alkalinity fields are obtained from the multivariate linear regression model LIAR (Carter et al., 2016; 2018) as a function of sea surface temperature and salinity, as well as climatological nitrate and dissolved silicate from World Ocean Atlas v2018[3]. Surface ocean pH, DIC, and the saturation state of sea water with respect to calcite and aragonite are calculated from reconstructed spco2 and talk, temperature, salinity, and nutrients by using the speciation software CO2sys (Van Heuven et al., 2011; Lewis and Wallace, 1998).

Uncertainty of pH, total alkalinity, DIC, and saturation state at each grid cell and time step is derived from LIAR and CO2sys error propagation (Orr et al. 2018). Inputs to the uncertainty propagation routine are default values for dissociation constants while they are space-time varying fields for other variables. Uncertainty estimates for talk and spco2 represent respectively systematic errors from LIAR (Carter et al., 2016; 2018) and randomness errors from the CMEMS-LSCE-FFNNv2 ensemble approach (Chau et al. 2022, 2024). For temperature and salinity, the uncertainties are defined as analysis errors associated with the CMEMS products (Table 2), and for nitrate, silicate, and phosphate climatologies the uncertainties are defined as 15% of the WOA18 data values[5].

https://www.ncei.noaa.gov/data/oceans/woa/WOA18/
https://archimer.ifremer.fr/doc/00651/76336/77327.pdf

III VALIDATION FRAMEWORK

This section describes the framework that was applied to conduct the validation process.

III.1 Surface ocean partial pressure of carbon dioxide

Only two thirds of the SOCAT target data are used for the training algorithm, leaving one third for model validation. The models are trained separately for each month, resulting in adaptive models with a common architecture but trained on different data. To increase the amount of data available for training, the models are trained using as target data from a 3-month moving window of the entire period from 1985 onward but excluding the reconstructed month.

Model output is assessed on independent SOCAT data and collocated reconstructed data. A leave-**p**-out cross-validation approach is applied (where **p** is the amount of data in the month considered for reconstruction) and 100 random subsamples of independent data are drawn from the SOCAT dataset. The network was run on each subsample. From these 100 results the mean was chosen as an estimate of monthly spco2 per grid cell.

All metrics are computed over the full reconstruction period and the period 1985-2021 for the comparison with the previous release. The assessment is broken down into the five oceanic basins used in data evaluation in the Regional Carbon Cycle Assessment and Processes project – phase 2[7] (Figure 3). Reconstructed data along the coast are evaluated within the SOCAT coastal zones defined within 400 km from the shoreline and collocated on the 0.25°× 0.25° SOCAT grid.

III.2 Surface ocean downward mass flux of carbon dioxide

There is no independent data for validating fgco2. CMEMS-LSCE-FFNNv2 estimates of air–sea CO_2 fluxes are assessed by comparison to published estimates derived from an ensemble of global ocean biogeochemical models (GOBMs) and an ensemble of data-based reconstruction models used in the Global Carbon Project (GCP, Friedlingstein et al., 2025). Note that in order to obtain estimates for the anthropogenic fluxes the data-based estimates need to be corrected for the preindustrial outgassing, i.e., of riverine carbon input, of 0.65 PgC/yr (Regnier et al., 2022).

https://github.com/RECCAP2-ocean/R2-shared-resources/tree/master/data/regions, last access 06/11/2024

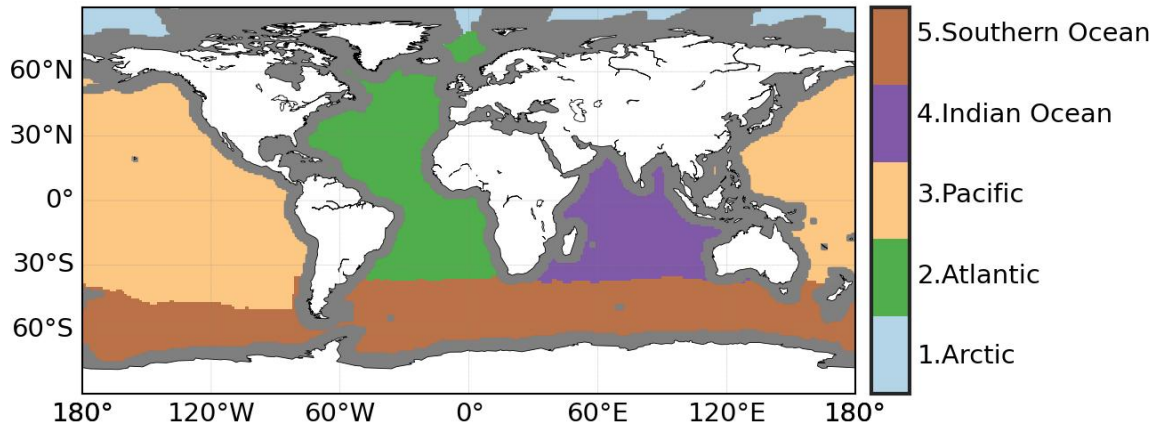

Figure 3: Map of the ocean used for model assessment. The coastal mask (grey) is consistent with the definition of coastal SOCAT coverage (400 km from shoreline; Bakker et al. 2016) collocated on 0.25° × 0.25° SOCAT grid.

III.3 Surface ocean pH, total alkalinity, dissolved inorganic carbon, and saturation state with respect to calcite and to aragonite

Total alkalinity, DIC, and pH from GLODAPv2.2023 bottle data are used for the evaluation of reconstructed surface ocean fields during 1985-2021. Only data from depths shallower than 10 m are used. Data are averaged per month and on a regular 0.25° × 0.25° grid, which resulted , respectively for talk, tco2 and ph, in ~ 14800,~ 16400 and ~ 9100 data points (25% for talk and tco2, 17% for pH of points distributed along the coastal sector). GLODAP pH data derived from the combination of talk and tco2 data are excluded from our product evaluation. The reconstruction metrics are calculated from co-located reconstructed values at these data points (Table 5 and **Error! Reference source not found.**).

IV VALIDATION RESULTS

The following subsections detail the results from the validation of the product.

IV.1 Surface ocean partial pressure of carbon dioxide

Skill scores computed for 5 oceanic regions are shown in Table 3. The table illustrates - for the full period - that the highest RMSDs (open: 32.03 µatm, coastal: 40.48 µatm) and MAD (open: 21.89 µatm, coastal: 28.05 µatm) are associated with the region having the lowest data density and high temporal and/or spatial pCO_2 variations (Arctic). The CMEMS-LSCE-FFNNv2 reconstruction over the coastal regions for the full period is roughly twice less effective than over the open ocean in terms of RMSD and MAD overall while it shows a rather good fit with $r^2 = 0.72$. The high RMSD reflects local high model errors along the continental shelves () which are characterized by complex physical and biological dynamics leading to high variability at small scales. Further model improvement is needed to capture such high spatial and temporal variability of surface ocean spco2 present in observations (see in Bakker et al. 2016; Laruelle et al., 2017, and references therein). An in-depth analysis on the model reconstruction over different oceanic basins for both open and coastal regions is presented in Chau et al. (2022, 2024b).

The average surface ocean spco2 for the period 1985-2024 is shown on **Error! Reference source not found.**a. High surface ocean spco2 values are associated to the upwelling of deep water, which is naturally enriched in DIC (Dissolved Inorganic Carbon, e.g., Equatorial Pacific upwelling). Low CO_2 partial pressures are found in cold northern and southern latitudes, a combination of the temperature effect on the solubility of CO_2 and the drawdown of DIC by biological activity. The spatial uncertainty of spco2 corresponding to the standard deviation of outputs from the 100 models from the CMEMS-LSCE-FFNNv2 is displayed on **Error! Reference source not found.**b. These maps illustrate the confidence level in the reconstruction of spco2. Large estimates of the uncertainties were found in the coastal regions, regions sparse or devoid of SOCAT data (see **Error! Reference source not found.**) such as Indian Ocean, Southern Ocean, and regions with high or low surface pressure of CO_2 such as East Pacific, Labrador Sea, North Western Pacific.

Table 3: Statistical validation for CMEMS-LSCE-FFNNv2 over the period 1985-2024. Comparison between reconstructed surface ocean spco2 and spco2 values from the SOCAT data that were not used during the algorithm training. Global statistics and regional examples (Figure 3), number of SOCATv2025 data per open (O) and coastal (C) ocean.

	Number of data		MAD (µatm)		RMSD (µatm)		r^2	
	(O)	(C)	(O)	(C)	(O)	(C)	(O)	(C)
Global	8202344	540404	9.45	16.47	13.89	26.69	0.84	0.72
Arctic	1912	34425	21.89	28.05	32.03	40.48	0.75	0.56
Atlantic	215092	276510	9.37	14.70	13.15	23.66	0.84	0.74
Pacific	474964	140706	9.49	17.42	14.30	29.21	0.85	0.69
Indian Ocean	18625	7104	7.45	12.89	10.24	23.65	0.89	0.70
Southern Ocean	109641	81659	9.57	16.26	13.53	24.71	0.73	0.65

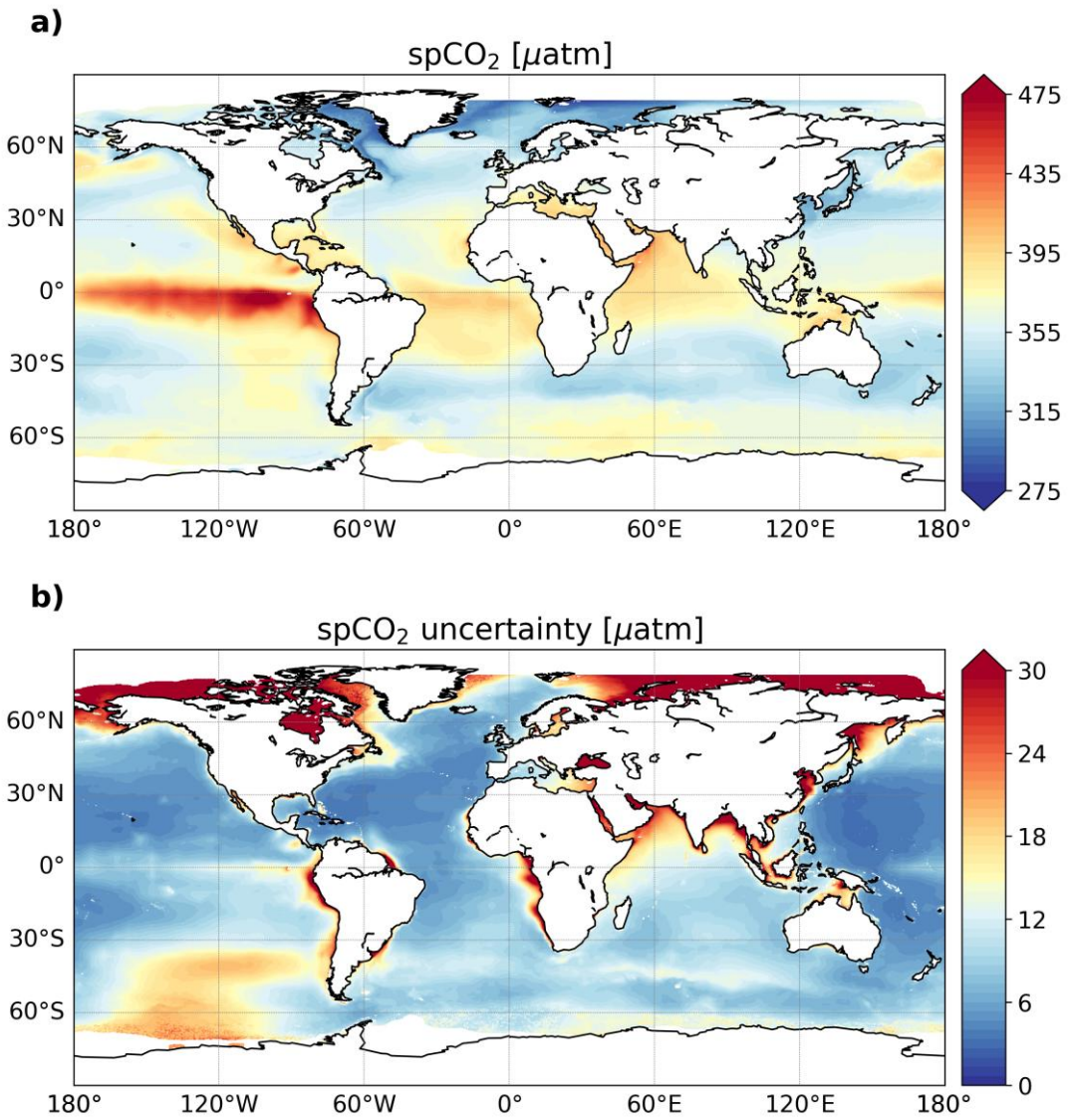

Figure 4: Reconstructed surface ocean carbon dioxide partial pressure averaged over the global ocean for 1985 to 2024. Mean (a) and standard deviation (b) of the 100-member CMEMS-LSCE-FFNNv2 model outputs are shown. The uncertainty estimate of the temporal mean per grid cell is computed following Section II.2.

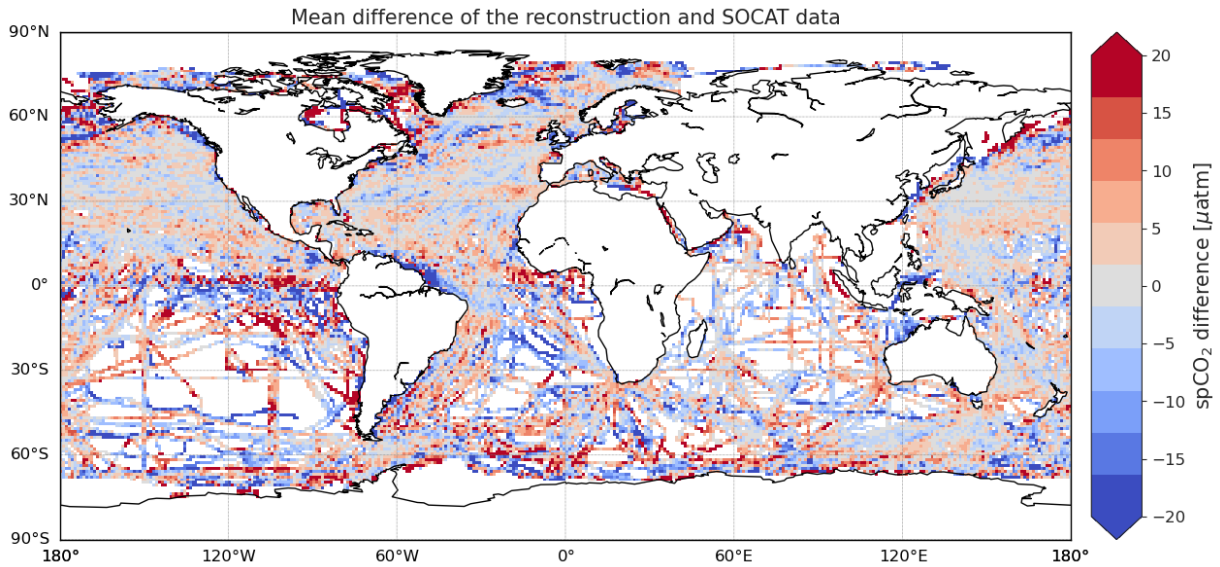

Figure 5: Map of the mean differences between MULTIOBS spco2 and the entire SOCAT dataset [µatm].

While the average bias of reconstructed surface ocean partial pressure compared to the SOCAT data is small, there are areas with persisting strong positive or negative biases (e.g., coastal seas and continental shelves, subpolar and polar regions, Eastern Pacific Ocean, Western South Atlantic). shows the full dataset of the differences from the observations for the period 1985-2024. It hides the still sparse data coverage for individual months, which challenges any reconstruction method. Even more problematic is the poor data coverage of large regions (e.g., coasts, high latitudes, Indian Ocean and South Pacific Ocean). Reconstructed monthly fields are very noisy with high uncertainty over areas with poor data coverage (see **Error! Reference source not found.**) and/or high variations of pCO2, and should be viewed with caution.

IV.2 Surface ocean downward mass flux of carbon dioxide

A comparison between CMEMS-LSCE and GCB2024 (Global Carbon Budget 2024, Friedlingstein et al. 2025) for the reconstruction of air-sea CO_2 fluxes is presented in Table 4. The GCB Ocean sinks are the average estimates from an ensemble of global ocean biogeochemical general circulation models (GOBMs) and an ensemble of data-based reconstruction methods.

The global ocean CO_2 sink corresponds to the global integral of fgco2 over space and time. The reconstructed field covers both the open ocean and coastal regions and hence approximately 96.37% of the total ocean area (361.9e6 km^2). In 2024 [resp. 2023], the global ocean CO_2 sink was 2.47±0.13 [resp. 2.58±0.13] PgC/yr (Figure 1). The average of yearly contemporary fluxes over the full period 1985-2024 is 1.68±0.10 PgC/yr with an interannual variability (temporal standard deviation) of 0.524 PgC/yr. Taking into account the total ocean area of 361.9e6 km^2 and the outgassing of river carbon of 0.65 PgC/yr yields an anthropogenic carbon uptake by the ocean of 3.12±0.32 PgC/yr [3.23±0.33 PgC/yr] for 2024 [2023] and 2.33±0.32 PgC/yr for the years 1985-2024. CMEMS-LSCE estimates of the anthropogenic uptake are of similar magnitude as the estimates by Friedlingstein et al. (2025) (see Table 4).

The ensemble-based approach yields spatio-temporal varying uncertainties which are consistent with uncertainty estimates of an ensemble of data-based and model-based approaches in Hauck et al. (2020). However, they are lower than the previous GCB published total estimate of ±0.4 PgC/yr. The latter estimate corresponds to the combination of random uncertainty (ensemble standard deviation of model- and data-based flux estimates) and systematic uncertainty (bias of model-based estimates in anthropogenic carbon cumulation).

Table 4: Air-sea CO_2 flux (PgC/yr). Comparison between CMEMS-LSCE-FFNNv2 and other estimates for periods in 1985-2023. The estimates of CMEMS-LSCE-FFNNv2 anthropogenic flux were computed by adjusting the contemporary flux estimates with the global ocean area of 361.9e6 km^2 and the riverine flux of 0.65 PgC/yr, as it is done in the GCB.

Methods		**Periods**			
		1990s	**2000s**	**2014-2023**	**2023**
CMEMS-LSCE-FFNNv2	Contemporary	1.40 ± 0.10	1.66 ± 0.09	2.21 ± 0.10	2.58 ± 0.13
	Anthropogenic	2.05 ± 0.32	2.31 ± 0.31	2.86 ± 0.32	3.23 ± 0.33
GCB2024		2.1 ± 0.4	2.3 ± 0.4	2.9 ± 0.4	2.9 ± 0.4

The map of average air-sea fluxes of CO_2 for the period 1985-2024 (**Error! Reference source not found.**) highlights the regional variability of fluxes over the global ocean and the coastal regions. Outgassing of CO_2 is associated with the upwelling of CO_2-rich subsurface waters (e.g., Equatorial Pacific). The northern and southern mid to high latitudes are sink regions. The North Atlantic stands out as a major area of CO_2 uptake explained by a marked strong cooling in winter and a vigorous phytoplankton bloom in spring and early summer. A region of enhanced uptake is also associated with the subtropical convergence zone in the southern hemisphere. Based on the gas exchange formulation, the uncertainties derived from the 100-member ensemble of fgco2 estimates can be interpreted as the scalar product of the spco2 uncertainties resulted from the CMEMS-LSCE-FFNNv2 and the gas transfer coefficients $k\rho L(1-f_{\text{sea ice}})$,

from Eq. 1. On the panel (b) of **Error! Reference source not found.**one notes that the uncertainty for the Southern Ocean, due to its strong and variable winds, is higher than for the other regions.

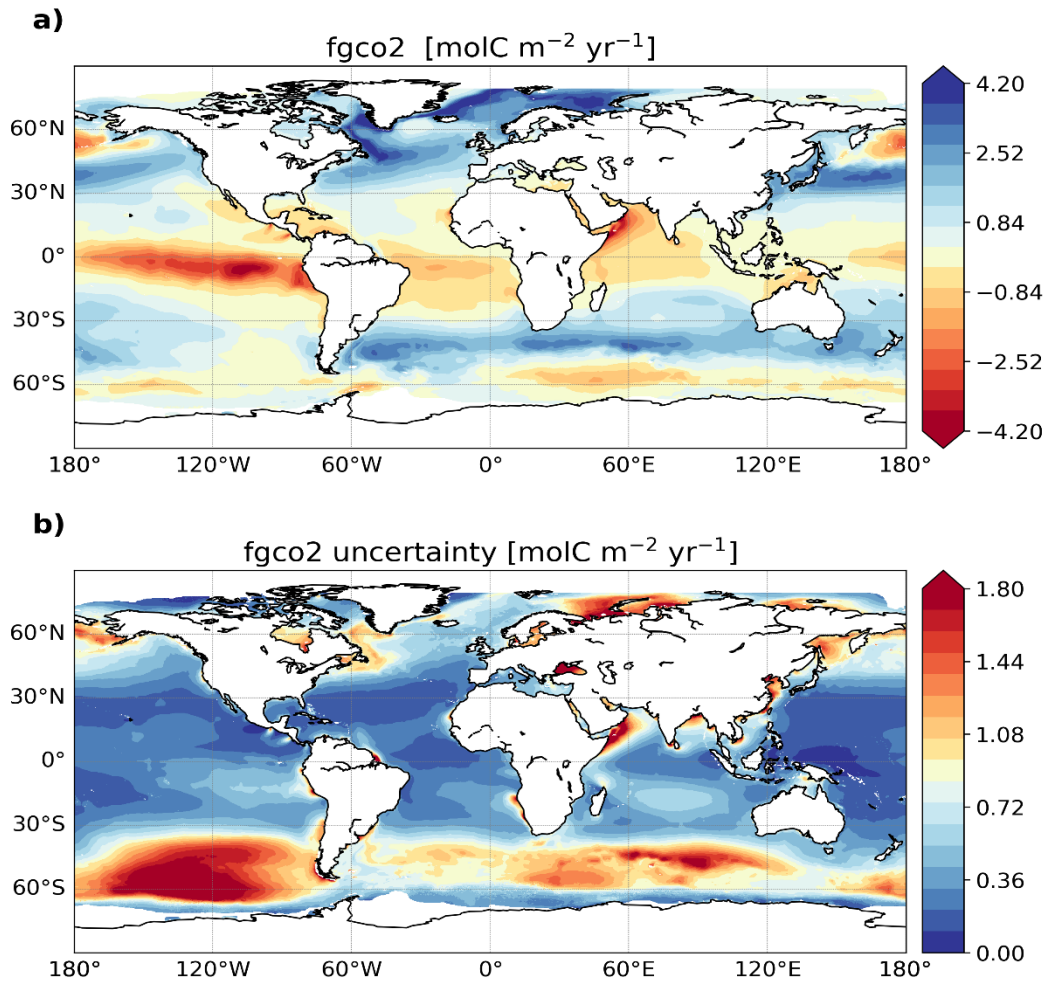

Figure 6: Reconstructed surface downward flux of carbon dioxide averaged over the global ocean for 1985 to 2024..Mean (a) (positive values correspond to ocean uptake) and standard deviation (b)computed from the 100-member ensemble of CMEMS-LSCE-FFNNv2 are shown. The uncertainty estimate of the temporal mean per grid cell is computed following Section II.2.

The global yearly integrated air-sea flux of CO_2 seen in Figure 1 is characterized by interannual variability in response to modes of natural climate variability, which occurs superposed on the trend driven by increasing atmospheric CO_2 levels. The Equatorial Pacific is a dominant source of CO_2 to the atmosphere. Equatorial Pacific outgassing is strongly modulated by ENSO dynamics and imprints global air-sea fluxes of CO_2 (Feely et al., 1999). During El Niño events (e.g., 1997-1998, 2015-2016), reduced upwelling translates into reduced outgassing of CO_2 and an enhanced global ocean net air-sea CO_2 flux. On the contrary, La Niña events (e.g. 2007-2008, 2020-2023) are characterized by increased upwelling, enhanced outgassing and a decreased global ocean net air-sea CO_2 flux (Chau et al. 2022 [**Error! Reference source not found.**]). The year 2023 was remarkable with large areas of the ocean experiencing strong positive SST anomalies co-occurring with an El Niño (2023/2024). The ocean CO_2 sink showed a strengthening in

2023. The sink was, however, smaller than expected from the sole relationship between El Niño and global ocean CO_2 uptake. The effect of El Niño (strengthening of the sink) was in part off-set by the excessive warming of extra-tropical waters (weakening of the sink), in particular the North Atlantic (Chau et al., 2024b; Müller et al., 2025). The alternation from El Niño to neutral conditions during 2024 left large portions of the ocean with high SST and resulted in a slightly weaker ocean C sink in 2024. The detailed analysis of underlying processes and dynamics needs further studies.

IV.3 Surface ocean pH, total alkalinity, dissolved inorganic carbon, and saturation state with respect to calcite and aragonite

In comparison with independent measurements from GLODAPv2.2023 over the period 1985-2021, the CMEMS-LSCE-FFNNv2 reconstruction results in open-ocean alkalinity [DIC] with a global MAD of 9.76 [11.37] µmol kg^{-1}, RMSD of 20.56 [21.12] µmol kg^{-1}, and r^2 value of 0.90 [0.92]. Large estimation deviation from GLODAP data is found in high latitudes and tropics. Coastal regions remain challenging for the extrapolation of these carbonate variables with much less model skill and higher uncertainty estimates than those corresponding to the open ocean (Table 5**Error! Reference source not found.** and **Error! Reference source not found.**9). However, these model errors and model uncertainties are below 10% of talk or of tco2 mean values. The appearance of spikes in the uncertainty maps of surface ocean alkalinity and DIC as well as of saturation state (**Error! Reference source not found.**) are highly correlated to that characterized by the uncertainty field of the CMEMS salinity product (Table 2[8]).

The assessment of the quality of the CMEMS-LSCE-FFNNv2 reconstruction with respect to pH data from the GLODAPv2.2023 bottle data set yields a global RMSD of 0.021 [0.037] pH and an absolute bias of 0.013 [0.023] pH units over the open [coastal] ocean. Model scores for the regional assessment are shown in Table 5 and Figure 9. CMEMS-LSCE produces reliable estimates of pH over the open ocean between 60°S and 60°N. Model errors (Figure 9 and uncertainty estimates) remain high over the coastal seas, continental shelves, and high latitudes. For a comparison with the existing global carbonate product (OceanSODA-ETHZ) over the open ocean, CMEMS-LSCE-FFNNv2 reproduces pH estimates with similar skill scores (Gregor et al. 2021 [Table 3]). This product qualification was based on both GLODAPv2019 and SOCCOM data (direct pH measurements collected by the Southern Ocean Carbon and Climate Observations and Modeling project).

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Table 5: Statistical validation for CMEMS-LSCE-FFNNv2 over the period 1985-2021. The assessment is based on the comparison between reconstructed surface ocean data, and GLODAPv2.2022 bottle data set for total alkalinity (talk), DIC (tco2), and pH over the period 1985-2021. Global statistics and regional examples (Figure 3), number of GLODAPv2.2023 data per open ocean (O) or coastal (C) region between brackets.

Basins	tco2 [µmol kg^{-1}]							
	Number of data		MAD		RMSD		r^2	
	(O)	(C)	(O)	(C)	(O)	(C)	(O)	(C)
Global	12342	4135	11.35	26.05	21.12	51.68	0.92	0.75
Arctic	57	465	90.38	81.65	108.57	123.32	0.78	0.32
Atlantic	3492	1941	12.64	21.25	28.66	38.08	0.71	0.70
Pacific	5547	763	10.68	16.71	15.58	26.98	0.93	0.69
Indian Ocean	1238	345	10.78	14.27	14.71	23.50	0.95	0.88
Southern Ocean	2008	621	9.19	17.44	13.26	25.74	0.91	0.77
Basins	**talk [µmol kg^{-1}]**							
	Number of data		**MAD**		**RMSD**		**r^2**	
	(O)	**(C)**	**(O)**	**(C)**	**(O)**	**(C)**	**(O)**	**(C)**
Global	11132	3738	9.76	27.23	20.56	56.57	0.90	0.83
Arctic	63	535	7.30	93.04	109.17	122.91	0.81	0.52
Atlantic	3779	1672	10.00	19.06	26.84	44.19	0.76	0.69
Pacific	4469	621	9.16	12.08	13.32	18.64	0.92	0.88
Indian Ocean	1235	360	10.76	12.37	14.82	16.40	0.92	0.93
Southern Ocean	1586	550	7.6	14.90	10.23	23.39	0.70	0.47
Basins	**ph[-]**							
	Number of data		**MAD**		**RMSD**		**r^2**	
	(O)	**(C)**	**(O)**	**(C)**	**(O)**	**(C)**	**(O)**	**(C)**
Global	7583	1548	0.013	0.023	0.021	0.037	0.71	0.51
Arctic	12	87	0.0.063	0.046	0.079	0.062	0.42	0.34
Atlantic	2583	819	0.013	0.023	0.023	0.038	0.64	0.48
Pacific	3613	463	0.013	0.023	0.018	0.036	0.75	0.54
Indian Ocean	575	68	0.011	0.010	0.024	0.013	0.64	0.81
Southern Ocean	800	111	0.014	0.017	0.019	0.024	0.69	0.63

Reconstructed surface ocean alkalinity varies spatially across and between ocean basins. Alkalinity is highest over the tropical Atlantic Ocean in line with high salinity concentration[9]. In the regions where salinity is reduced by precipitation and/or seasonal sea-ice melting (e.g. subpolar Pacific Ocean, Arctic River outflows, Equatorial band, Eastern Indian Ocean and Indonesian Archipelago), surface ocean

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alkalinity is low. High levels of DIC are reconstructed throughout the Southern Ocean because of upwelling of CO_2 rich waters and subsequent DIC-spreading by prevailing westerlies. Outside of upwelling areas, DIC levels are modulated by biological draw-down and temperature control on CO_2 solubility regionally resulting in pronounced seasonal cycles (e.g. subpolar regions, Feely et al. 2001). Uncertainty estimates of the reconstructed alkalinity and DIC are below 5% of the climatological mean values over the open ocean.

Reconstructed saturation states (Ω) with respect to calcite and aragonite (polymorphs of the mineral calcium carbonate, $CaCO_3$) show similar spatial variability across the ocean basins (**Error! Reference source not found.**). High values of saturation state with respect to calcite and aragonite are reconstructed over the subtropical and tropical bands (between 30°S and 30°N) reflecting relatively high ALK over DIC ratios. A decrease in the saturation state is found in high latitudes and upwelling regions. Lowest values are associated with low alkalinity and high DIC in high latitude waters. While surface waters remain in average oversaturated with respect to calcite and aragonite (Ω = 1), saturation states below 1.5 are reached locally. Uncertainty estimates for surface ocean saturation state are less than 10% of its global mean, exceptionally for the coastal regions (**Error! Reference source not found.**).

Reconstructed surface ocean pH is given on total H^+ scale. Low pH values are found in upwelling regions (e.g., Equatorial Pacific, Arabian Sea) where dissolved inorganic carbon (DIC) enriched subsurface waters reach the surface ocean. Higher values are found in subpolar and polar waters. The uncertainty map of pH displayed on the panel b) t is derived through a monthly uncertainty propagation for the 100-ensemble mean and ensemble spread of spco2 following Orr et al. (2018) and the speciation software CO2sys (Lewis and Wallace, 1998; Van Heuven et al., 2011) (see Sec. II.2 (c) for details). Estimates of spco2 and associated uncertainties (see **Error! Reference source not found.**) have the largest impact on the estimation of pH uncertainties.

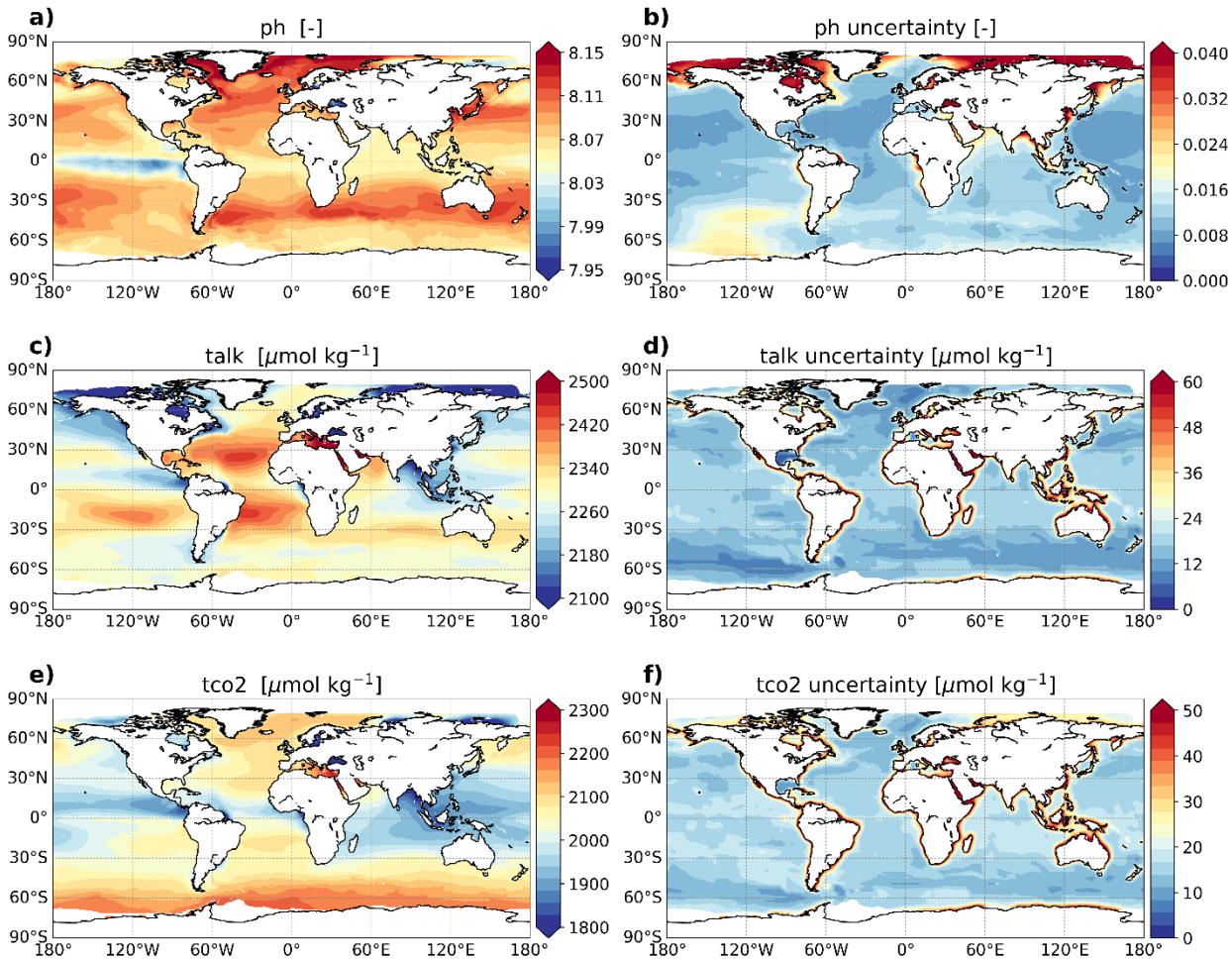

Figure 7: Reconstructed surface ocean pH, total alkalinity, and dissolved inorganic carbon (DIC) averaged over the global ocean for the period 1985–2023. Panels (a,b) show the temporal mean and associated uncertainty (1σ) for pH, panels (c,d) for total alkalinity, and panels (e,f) for DIC. Temporal mean estimate and uncertainty are computed for the 100-member ensemble of CMEMS-LSCE-FFNNv2. The uncertainty estimate per grid cell is computed following Section II.2. Spikes in the uncertainty maps of surface ocean alkalinity and DIC are traced back to the uncertainty field of the CMEMS salinity product.

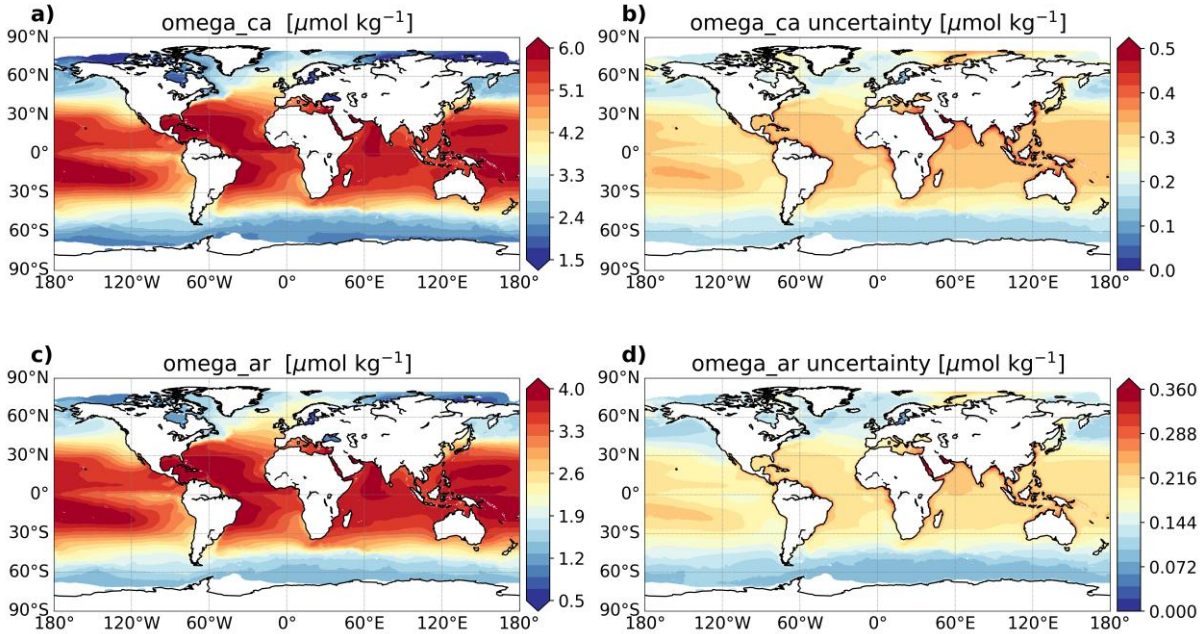

Figure 8:Reconstructed surface ocean saturation state with respect to calcite (omega_ca) and aragonite (omega_ar) averaged over the global ocean for 1985 to 20232024. Panels (a, b) show the temporal mean and associated uncertainty (1σ) for ocean saturation state with respect to calcite and panels (c, d) with respect to aragonite. Temporal mean estimate and uncertainty are computed for the 100-member ensemble of CMEMS-LSCE-FFNNv2. The uncertainty estimate per grid cell (1σ) is computed following Section II.2.. Spikes in the uncertainty maps of surface ocean saturation state are traced back to the uncertainty field of the CMEMS salinity product (Table 2, [12]. [13]).

[12] Copernicus Marine Data visualization tool for sea surface salinity error
[13] Copernicus Marine Data visualization tool for sea surface salinity error

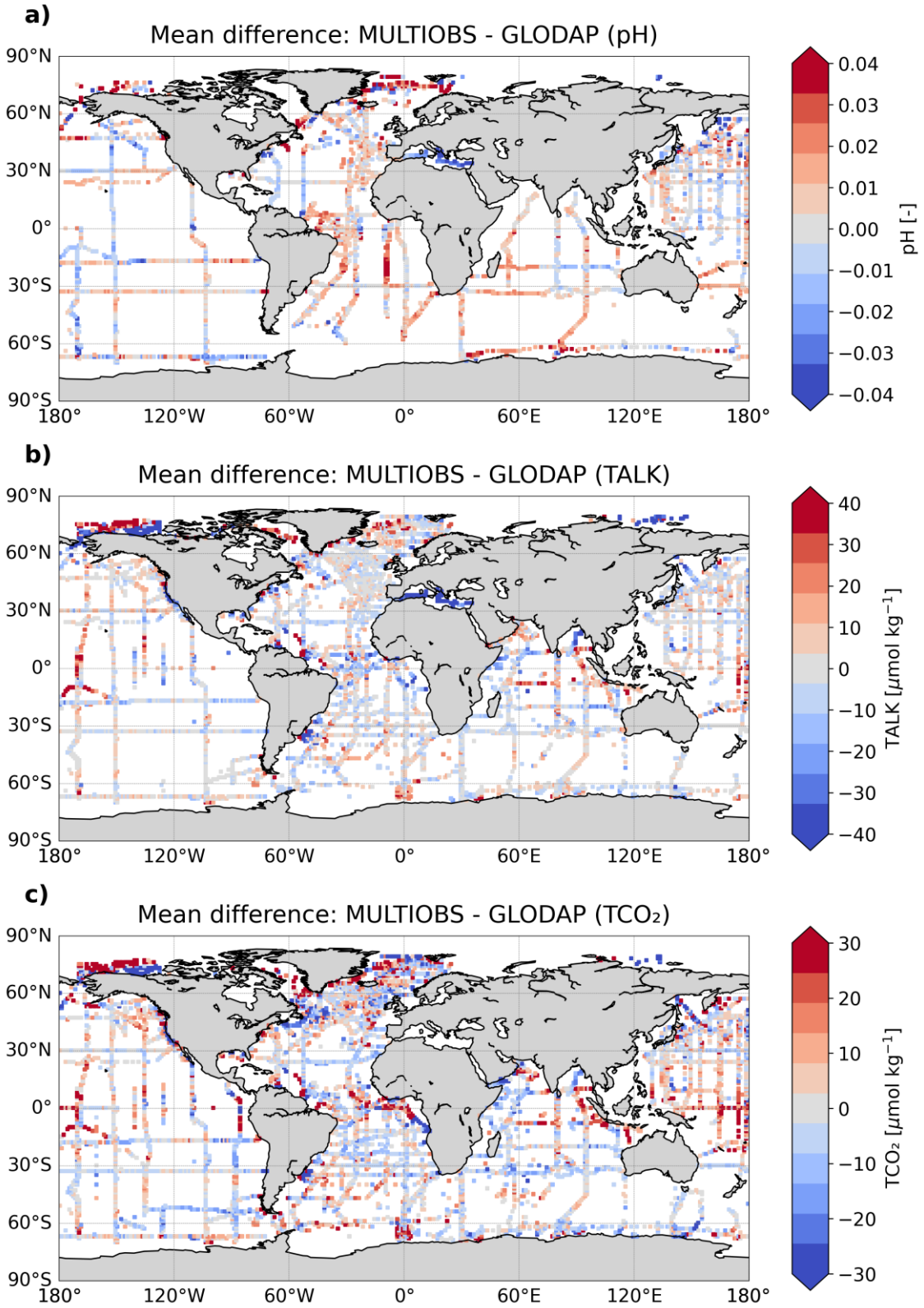

Figure 9: Maps of the mean of the differences between MULTIOBS and GLODAPv2.2023 bottle dataset for pH (ph)(a), total alkalinity (talk)[b]) and DIC (tco2)(c).

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 is the new name of MULTI-OBS_GLO_BIO_CARBON_SURFACE_REP_015_008.

The first release, in 2020, of the MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008this product corresponded to a reprocessed and extended time series of monthly spco2, fgco2, and ph over the global ocean at 1° spatial resolution. Estimates of spco2 were obtained from an ensemble of feed forward neural networks (FFNNs) (Chau et al. 2022). The system is referred to as CMEMS-LSCE-FFNNv2 and is an improvement (method-wise) from the 1st version (LSCE-FFNN-v1) which followed the principles laid down in Denvil-Sommer et al. (2019). See Chau et al. (2020) for intercomparison. All variables were distributed with associated uncertainties derived from the 100-member ensemble.

Since the year 2021, the full CMEMS-LSCE-FFNN product including data over the coasts and high latitudes (above 60°N) has been submitted to CMEMS. Due to these changes, all statistics reported in the current document are significantly different from those in the 2020 version.. Note that the larger errors are derived from the reconstruction over the coasts and high latitudes.

Since the year 2022, time series of the 4 new variables (surface ocean alkalinity, dissolved inorganic carbon, saturation state with respect to calcite and aragonite) and associated reconstruction uncertainties have been provided. See Section IVfor further updates of the evaluation corresponding to this release.

The year 2023marked the first release of CMEMS-LSCE datasets of the seven carbonate system variables at monthly, 0.25° resolutions. A full data assessment is presented in Chau et al. (2024).

In 2024, the product name has changed from MULTIOBS_GLO_BIO_CARBON_SURFACE_REP_015_008 to MULTIOBS_GLO_BIO_CARBON_SURFACE_MYNRT_015_008 and now to MULTIOBS_GLO_CARBON_SURFACE_MYNRT_015_008. It contains two datasets **cmems_obs-mob_glo_bgc-car_my_na_irr-i** and **cmems_obs-mob_glo_bgc-car_nrt_na_irr-i** . This second dataset was created within this product in order to cover the recent months for spco2 and fgco2, after the last month of the SOCAT database. This dataset is extended every month for the previous month using the FFNNs of the last SOCAT year.

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

This new release CMEMS-LSCE (v2025) is mainly a straight-forward update and extension by one year of the previous version (v2024) using the latest SOCAT release, SOCATv2025. It also covers the months after the last month of SOCATv2025 for spco2 and fgco2 (only), with a lag of 1 month in "prediction mode" (**Error! Reference source not found.**). This mode is a natural extension of the "reconstruction mode" which only considers years within the SOCAT database. In prediction mode, the neural networks trained with the last year of SOCATv2025 are reused. As described in Chau et al. (2024a), a retrospective prediction showed only a modest degradation of the quality of spco2 and fgco2 compared to the reference reconstruction. The prediction maps can therefore be used with the same confidence as the reconstruction maps.

Error! Reference source not found. illustrates the consistency between versions v2024 and v2025 with the monthly global mean of pH and the monthly integration of air-sea fluxes of CO_2.

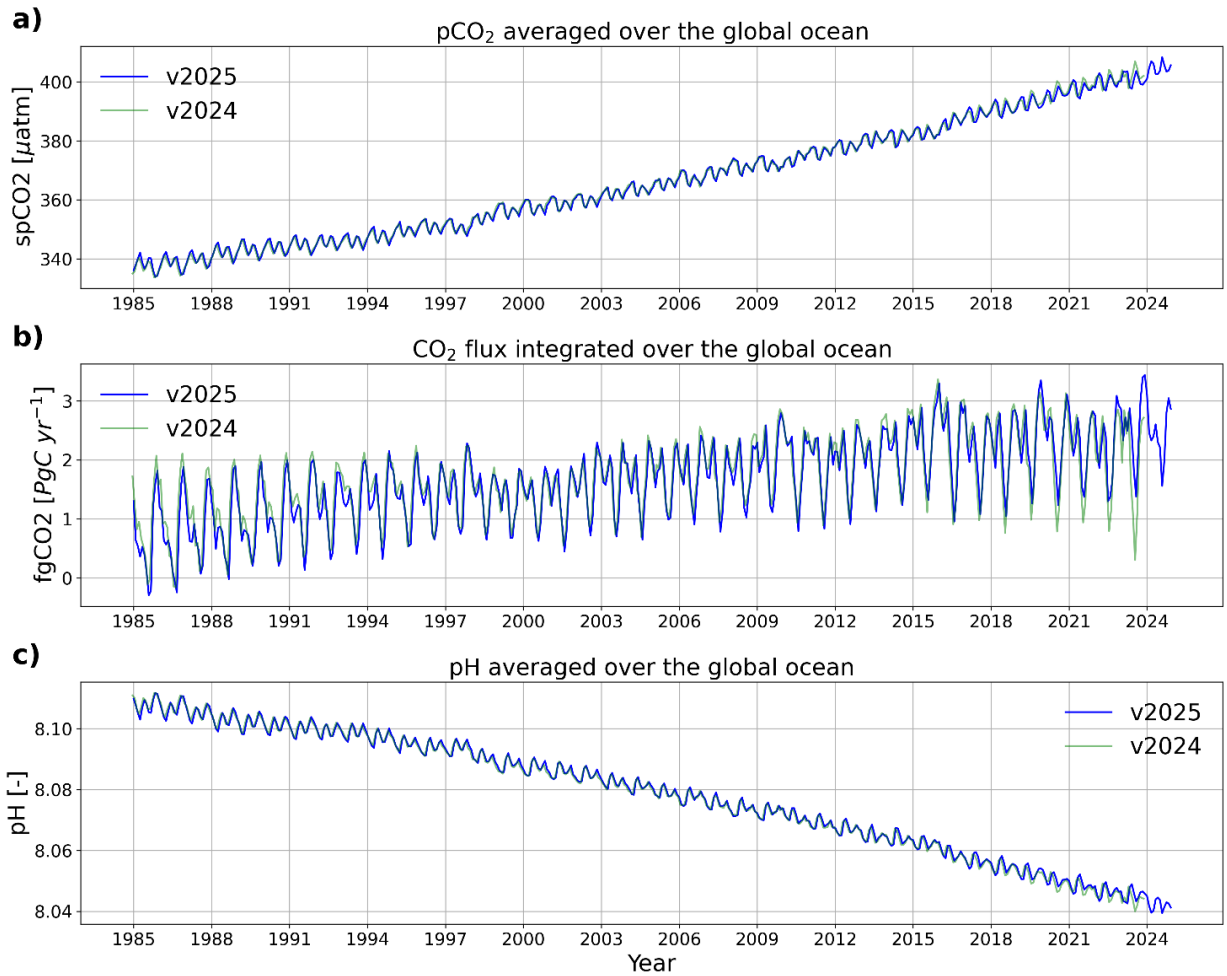

Figure 10: Monthly integration of surface ocean partial pressure of carbon dioxide spCO2 (a), air-sea fluxes fgco2 (b) and global mean monthly pH (c) estimated by CMEMS-LSCE-FFNNv2 (2024, 2025).

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